PREP 4, LESSONS 7–8

Using radiation and half-life

Ionising power, range and penetration; choosing a source; half-life from a graph

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons, 1 mark each

1 State the standard unit of activity. DN 262 [1]

Becquerel (Bq)

2 Describe how the mass number and atomic number change when an alpha particle is emitted. DN 280 [1]

Mass number decreases by 4. Atomic number decreases by 2.

3 State what a beta particle is. DN 268 [1]

A high-speed electron from the nucleus

Key ideas LESSONS 7–8

Show your working

4 The most ionising radiation is **alpha** | beta | gamma [1]

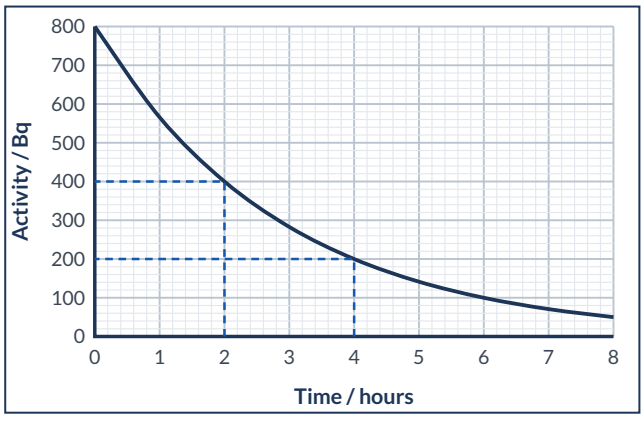
5 A factory uses a radioactive source and a detector to check the thickness of paper.

Draw a ring around the radiation it should use: alpha **beta** gamma Explain your choice. [2]

Alpha would all be absorbed and gamma would all pass through ✓

beta is partly absorbed, so the count-rate changes when the thickness changes ✓

6 The graph shows how the activity of a source changes with time.



a Use the graph to find the half-life. Show on the graph how you did it. [2]

Read from 800 to 400 Bq (half) ✓ half-life = 2 hours ✓

b Define half-life. [1]

The time for the number of unstable nuclei (or the activity) to halve ✓

c Radioactive decay is random. Explain what this means. [1]

You cannot predict which nucleus will decay next, or when ✓